

A Tutorial in Dynamic Systems Simulation and Modelling With R, ctsem, and lme4: Couples' Affect Dynamics Over Time

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Understanding how people change is a fundamental goal of psychological science. However, translating complex ideas about psychological dynamics into formal models can be challenging without the right tools. In this tutorial, we introduce a approaches using R and the ctsem package to help researchers build and understand dynamic systems models that capture some of the complexity of psychological processes. We emphasize iterative model-building through simulations, model fitting, and visualizations of the model-implied dynamics alongside their fit to data. We begin with familiar linear models in lme4 and gradually transition to ctsem, which allows us to incorporate complexities such as state-dependent change, random fluctuation distinct from measurement error, covariate effects, interactions, and external inputs. These modelling concepts are illustrated using a running example of affect dynamics in couples therapy, demonstrating how key conceptual and methodological ideas in dynamic systems modelling come together. Our aim is to provide a general framework for understanding dynamic systems modelling and to encourage further exploration of theory-driven statistical approaches in psychological research.

Impact Statement

This tutorial shows how psychological researchers can use R and the ctsem package to build and understand dynamic systems models that capture the complexity of psychological processes. By providing a structured approach to model building, we aim to facilitate the translation of theoretical concepts into empirical models, ultimately enhancing our understanding of psychological concepts and their interrelations over time.

Keywords: dynamic systems, continuous time, hierarchical modelling, longitudinal data analysis, affect dynamics, ctsem, lme4, state-space models

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